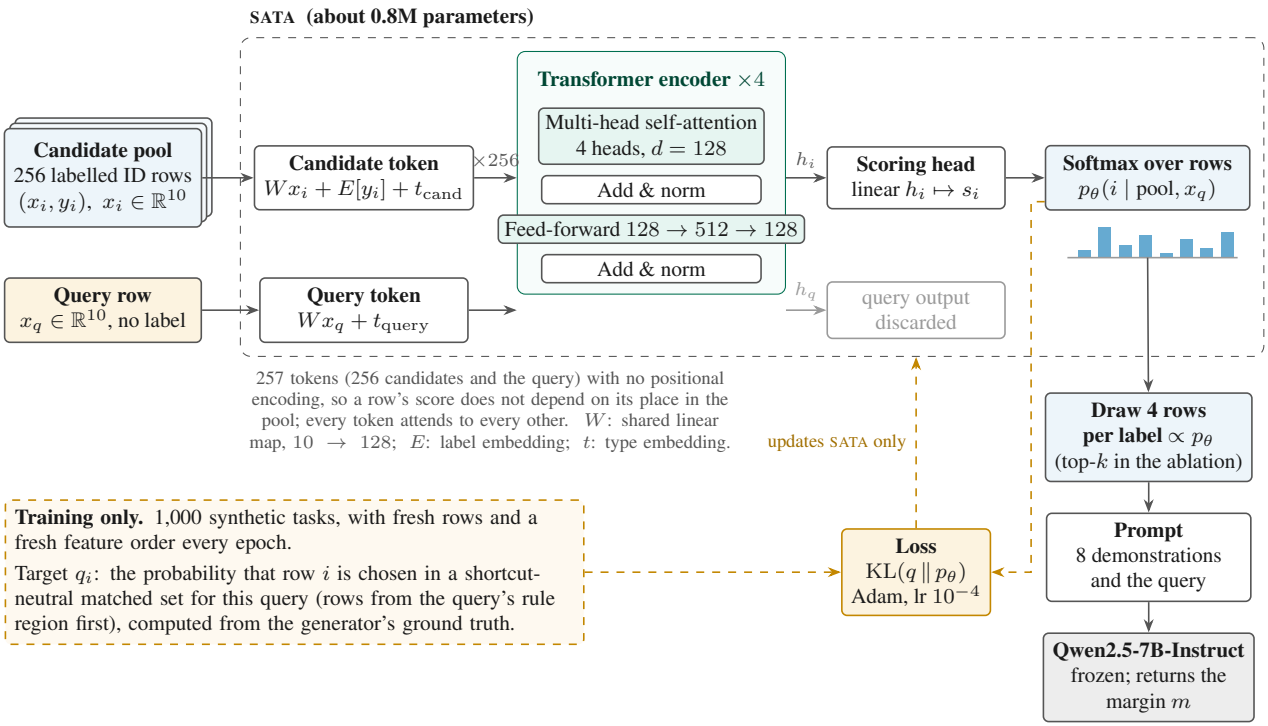

SATA (about 0.8M parameters)
Candidate pool
256 labelled ID rows
(x_i, y_i), $x_i \in \mathbb{R}^{10}$
Candidate token
$Wx_i + E[y_i] + t_{\text{cand}}$
×256
Transformer encoder ×4
Multi-head self-attention
4 heads, $d = 128$
Add & norm
Feed-forward 128 → 512 → 128
Add & norm
h_i
Scoring head
linear $h_i \mapsto s_i$
Softmax over rows
$p_\theta(i \mid \text{pool}, x_q)$
Query row
$x_q \in \mathbb{R}^{10}$, no label
Query token
$Wx_q + t_{\text{query}}$
h_q
query output
discarded
257 tokens (256 candidates and the query) with no positional encoding, so a row's score does not depend on its place in the pool; every token attends to every other. W: shared linear map, 10 → 128; E: label embedding; t: type embedding.
updates SATA only
Draw 4 rows
per label ∝ p_θ
(top-k in the ablation)
Prompt
8 demonstrations
and the query
Qwen2.5-7B-Instruct
frozen; returns the
margin m
Training only. 1,000 synthetic tasks, with fresh rows and a fresh feature order every epoch.
Target q_i: the probability that row i is chosen in a shortcut-neutral matched set for this query (rows from the query's rule region first), computed from the generator's ground truth.
Loss
$\text{KL}(q \,\|\, p_\theta)$
Adam, lr 10^{-4}